[INCH-POUND]
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COMMERCIAL ITEM DESCRIPTION

TEST SET, RADIO FREQUENCY POWER (50 MHz to 26.5 GHz)

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE.

This CID describes the minimum requirements for a Radio Frequency Power Test Set which consists of a peak power meter, sensor(s), and applicable accessories. The test set, as described in this CID, shall have a frequency range of 50 MHz to 26.5 GHz and will be used for measuring peak power and continuous wave power over selected frequencies.

2. CLASSIFICATION.

This document outlines requirements for both single and dual channel equipment. The determination as to whether the peak power meter will be a single or dual channel instrument will be specified at the time of procurement.

2.1 Type I - a single-channel peak power meter with appropriate sensor(s).

2.2 Type II - a dual-channel peak power meter with appropriate sensor(s).

3. SALIENT CHARACTERISTICS.

3.1 Safety and Environmental. The test set shall be solid state and enclosed to provide protection from electrical shock. This test set will be used on a bench top, as a rackmount, and occasionally as a portable test set in widely diverse environments.

Beneficial comments, recommendations, additions, deletions, clarifications etc., and any data which may improve this document should be sent to: Technology & Industrial Services Division, SA- ALC/TILDD, 306 Tinker Drive, Building 207, Kelly AFB, TX 78241-5916, by using the Standardization Document Improvement Proposal (DD Form 1426) included with this document, or a letter response.

AMSC N/A FSC 6625

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3.2 Electrical Power Sources. This test set shall operate from nominal commercial, military and shipboard power sources of 110/220 volts AC (± 10%) at line frequencies of 50 to 60 Hz. Power consumption shall not exceed 250 VA.

3.3 Frequency Range. The minimum frequency range of the peak power meter with sensors shall be 50 MHz to 26.5 GHz.

3.4 Power Meter Range. The peak power meter shall have, as a minimum, a range of minus 20 dBm to plus 15 dBm in the pulse and continuous wave modes.

3.5 Sensor Range. The sensor(s) shall have, as a minimum, a power range of minus 20 dBm to plus 15 dBm in the pulse and continuous wave modes.

3.6 Pre-conditioning requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn-in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

3.7 Calibration. The peak power meter shall be front-panel controlled. The method of calibration may be front-panel adjustment or front-panel automatic calibration. The calibration interval shall be a period of one year or greater based on an operating time of 2,000 hours.

3.8 Power Reference Calibrator. The power calibration source shall be an internal 1.0 GHz oscillator with a Type N connector on front panel or an internal 50 MHz oscillator with a Type N connector on the front panel.

3.8.1 Accuracy. The power reference accuracy shall be plus or minus 2.5% at 1 milliwatt.

3.9 Remote Operation. The peak power meter shall have an IEEE-488 interface for remote set up, control, and measurement operation.

3.10 Self-Test. The test set shall have a built-in self-test to fault isolate the cause of errors.

3.11 Display. The power meter shall have a digital display for all functions and contain a display which provides an analog representation of the pulse being measured. Additionally, the display shall provide, as a minimum, information on the amplitude and timing of the measured pulse.

3.12 Peak /Continuous Wave (CW) Selection. A front panel control shall allow the selection of either the peak or CW modes of measurements.

3.13 Interconnect cable. The power meter shall come with an interconnect cable for connecting the sensor to the instrument. The sensor cable shall be detachable from the sensor head and the cable shall be a minimum of 5 feet.

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3.14 RF Connection. The sensor RF interconnects shall be Type N or Type K.

3.15 Mean Time Between Failures (MTBF). The equipment MTBF shall be 2,000 hours. The lower test MTBF shall be 2,000 hours and the upper test MTBF shall be 4,000 hours.

3.16 Rackmount. Rackmount capability shall be available as an option for each test set.

3.17 Tilt Handle. A tilt handle will be available as an option.

3.18 Dimensions and Weight. The test set's component dimensions and weight shall be consistent with current commercial offerings such that:

3.18.1 each peak power meter shall be no larger than 8.0 inches high, 18.0 inches wide, and 22.5 inches deep;

3.18.2 total weight, including all accessories (excluding manuals) shall not exceed 49 lbs.

3.19 Temperature Requirements.

3.19.1 Operating Temperature. The test set shall meet performance and accuracy requirements in an operating environment of 0 to plus 50 degrees Celsius.

3.19.2 Non-operating Temperature. The test set shall meet performance and accuracy requirements after storage in an environment of minus 10 degrees to plus 60 degrees Celsius.

3.20 Manuals. Each test set shall be delivered with a user's manual that contains information on the operation and maintenance of the equipment. The manual shall include a parts breakout or a parts list.

4. REGULATORY REQUIREMENTS.

In accordance with Section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recoverable materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

5. QUALITY ASSURANCE PROVISION.

Product Conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

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6. PACKAGING.

The test set shall be packaged in accordance with normal commercial practices and packed to assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage. The shipping shall be in compliance with national motor freight classification and uniform freight classification. Shipping containers, markings, and preservation shall be as specified in the contract or order.

7. NOTES.

7.1 Metrics Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest version of Federal Standard No. 376, and all other requirements of this CID are met.

7.2 Federal and military documents are available from: Defense Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094; Phone: 215/697-2179.

7.3 Known Acceptable Products. The following products are known to have previously met the requirements of this CID and need not be resubmitted for bid samples or certification. All other interested manufacturers must meet the requirements of this CID and submit to bid testing and certification as required.

Manufacturer	Model	Type
Boonton Electronics	4400 one-channel meter with 56218 and 56526 sensors **or** 56218-S/2 sensor	I
	4400-01 two-channel meter with 56218 and 56526 sensors **or** 56218-S/2 sensor	II
Gigatronics, Inc.	8501 one-channel meter with 17267 sensor	I
	8502 two-channel meter with 17267 sensor	II
Hewlett Packard Co.	8991A-001 one-channel meter, 84813A & 84815A sensors	I
	8991A two-channel meter with 84813A & 84815A sensors	II

MILITARY INTERESTS:

Custodians:
Air Force - 99
Army - MI
Navy - SH

Review Activities:
Army - MI
Navy - SH
DLA - ES

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

Preparing Activity:
Air Force - 82

Project: 6625-0810

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